

Project & Data Management Plan

“Training Models on Moral Dilemma: How Artificial Intelligence (AI) “thinks” about Morality”

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Abstract—Project plan and outline in order to Evaluate Moral Reasoning and cognition in AI

I. PROJECT OVERVIEW

A. Background and Related Research

Large Language models tend to exhibit moral reasoning through reasoning patterns that it is trained upon, representing superficial understanding. Due to its natural reliance on statistical patterns, it lacks proper in-depth understanding of ethical considerations and principles.

This exploration takes from John Searle’s Chinese Room Argument (Searle, 1980), which stands to contest the assumption that genuine moral perception/cognition (semantics) versus imitation of pattern recognition through statistical association (syntax).

Modern investigations with goals of instilling ethical reasoning in AI was attempts with The Scruples data sets (Lourie, et al., 2021), with attempts to observe whether models were able to learn moral acceptability based on public/social consensus.

Similar grounded approach was further delved by (Hendrycks, et al., 2021) through the ETHICS dataset. Introducing significantly more structure with moral reasonings based on established philosophical frameworks like deontology, and utilitarianism.

Which there after leads to the, The Delphi model (Jiang, et al., 2021), on foundations from Hendrycks, was one of the pioneering large-scale attempts to build and AI system with offering not merely yes or no classifications (that was implemented in the hedrycks datasets) but also including and addressing open-ended responses.

These questions tackle the reliability for models to be able to replicate accurately moral concepts. However, lacking the ability to look under the hood of the trained model and know whether there is truly semantic understanding of morality or mere syntactic mimicry through statistical means.

B. Research Question & Objectives

This paper foremost aims to explore **how machine learning models performance across absolute and relative morality**, where absolute ethics represent a fixed normative standard (i.e. deontology), relative ethics capture moral judgements that allow fluidity (i.e. utilitarianism,

emotivism). These comparisons give rise to whether models are able to generalise across differing frameworks or are only able to obey specific types of moral paradigms.

Furthermore, the study examines **the consistency of AI ethical judgments under different frameworks of similar dilemmas**. By rewriting and testing the stability of responses in changed contexts.

Extending each individual lines of questioning, the overarching aim is to understand, **how much these systems learn and understand on a principled level, as opposed to merely imitating surface forms of language** – a distinction that is motivated in part by John Searle’s Chinese Room Argument. (Searle, 1980)

C. Main Contributions

This paper seeks to contributes both conceptually and practically. In the emerging field of artificial moral reasoning and AI transparency. Conceptually, it provides a framework for evaluating, through combining philosophical discourse with empirical model testing, AI systems demonstrate genuine moral understanding or merely mimic moral reasoning through statistical pattern recognition.

From a practical perspective, the project aims to reach measurable criteria and methods for assessing AI ethical reasoning. This includes assessment metrics for ethical consistency and justification quality, and exploring methods to examine how models internally encode ethical concepts. Through this examination, can inform the design of transparent and accountable AI systems that are able to align more closely with human ethical expectations.

Concrete results of this research include:

- A prototype evaluation framework to benchmark AI ethical reasoning across absolute and relative ethical frameworks.
- Empirical findings demonstrate how different training regimes (theory-based vs. imitation-based) affect moral generalization and reasoning consistency.
- Utilizing interpretability techniques to distinguish genuine understanding from statistical imitation – such as moral coherence, causal reasoning and counterfactual robustness.
- Recommendations for incorporating ethical reasoning modules into AI governance, education, and policy evaluation systems.

II. PROJECT PLAN

A. Task list:

STAGE	TASKS	START	END	NO. DAYS
	Finalize research question and scope	21-9-25	4-10-25	10

Project Scoping & Background Research	Conduct literature review on AI + morality, datasets	28-9-25	19-10-25	15
	Literature review on model selection	28-9-25	19-10-25	15
	Define evaluation metrics and success criteria	14-10-25	24-10-25	8
Dataset Design and Cleanup	Choose dataset focus (cross-cultural, uncertainty, etc.)	10-10-25	30-10-25	14
	Combining Dataset	29-10-25	31-10-25	2
	Cleaning up Data set	31-10-25	16-11-25	10
	Preliminary data set exploration	16-11-25	23-11-25	5
Model Selection & setup	Choose baseline models (rule-based, shallow)	23-11-25	27-11-25	4
	Choose transformer models or other NLPs	27-11-25	8-12-25	7
	Set up training environment (Python, GPUs, libraries)	3-12-25	4-12-25	1
	Model Training	4-12-25	22-1-26	35
	Fine-tune models on custom dataset	22-1-26	16-3-26	37
	Sanity-check and validate model outputs	11-3-26	20-3-26	7
Evaluation	Define and calculate evaluation metrics (accuracy, robustness, bias)	20-3-26	24-3-26	2
	Conduct interpretability analysis	24-3-26	13-4-26	14
	Cross-cultural or moral uncertainty tests	29-3-26	18-4-26	15
Analysis & Reflection	Relate findings to philosophical framing	13-4-26	24-4-26	9
	Identify limitations, biases, implications	20-4-26	29-4-26	7
Write-up & Presentation Prep	Draft report/paper (intro, method, results)	3-4-26	29-4-26	18
	Create figures, charts (attention maps, embeddings)	13-4-26	4-5-26	15
	Revise, finalize, and prepare presentation	29-4-26	19-5-26	14

Fig. 1 List of task and key milestones with estimated allocated time

B. Timeline:

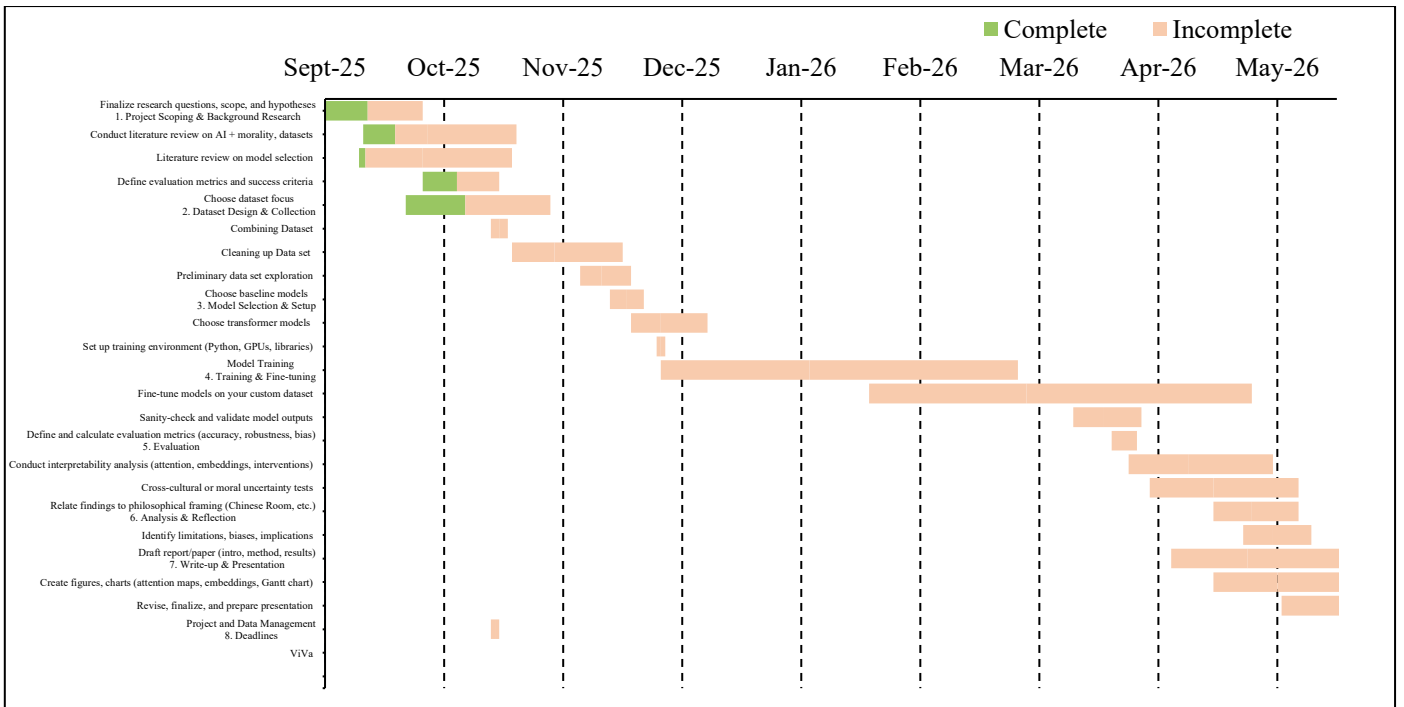


Fig. 2 Overview of Gantt outlining project timeline on 30th Oct 2025

From the two figures above illustrating the overall project task list and timeline, significant bulk of time has been mainly allocated to two areas, firstly data preprocessing and model training and finetuning.

III. DATA MANAGEMENT PLAN

A. Chosen Datasets

DATASET	LINK
ETHICS	https://github.com/hendrycks/ethics (Hendrycks, et al., 2021)

MoRe Bench

<https://github.com/morebench/morebench>
(Chiu, et al., 2025)

ETHICS was chosen for its pioneering grounding design encapsulating many different ethical frameworks. However, its annotation and classification are binary which is where, though on the flip side of the coin, the **MoReBench** was likewise selected for its pluralistic

moral reasoning, and incorporating human-annotated explanation. the MoReBench dataset allows for more in-depth evaluation.

The MoReBench dataset shares similar moral frameworks as well allowing both datasets to be concatenated together for more dimension in analysis.

1) ETHICS Dataset

a) Overview

The ETHICS dataset was one of the first large-scale, structured attempts to introduce philosophical rigor into moral reasoning datasets.

b) Metadata & Structure:

ATTRIBUTE	DESCRIPTION
Data Format	.json, .csv, .txt
Size	~33 MB
Records	~130,000 text scenarios
Annotation	Binary classification (acceptable = 1, unacceptable = 0)
Moral Subdomains	Deontology, Utilitarianism, Virtue Ethics, Commonsense Morality, Justice
Source / Origin	Created by UC Berkeley, UCSD, and Google Research (ICLR 2021)
Repository	https://github.com/hendrycks/ethics
Data Splits	Pre-divided into training, validation, and test sets

Fig. 3 Table metadata descript of ETHICS dataset

2) MoRe Bench Dataset

a) Overview

MoReBench represent more recent attempts (having been published in 2025) for a comprehensive large-scale effort to access procedural and pluralistic moral reasoning in models.

The interdisciplinary team in UC Berkeley, Stanford University seeks to test how well AI systems can justify morality across diverse moral frameworks beyond binary outcomes.

b) Metadata & Structure:

ATTRIBUTE	DESCRIPTION
Data Format	.json, .csv, .txt
Size	~200 MB
Records	~ 50 000 human-authored moral scenarios and justification pairs
Annotation	Multi-labelled open-ended annotations with reasoning
Moral Subdomains	Deontology, Utilitarianism, Virtue Ethics, Justice, Care ethics
Source / Origin	Created by UC Berkeley, UCSD, and Google Research (ICLR 2021)
Repository	https://github.com/morebench/morebench
Data Splits	Pre-divided into training, validation, and test sets

Fig. 4 Table metadata descript of ETHICS dataset
c) Datasets Ethical Compliance:

CRITERION	ETHICS	MoRE BENCH
GDPR Compliance	Dataset is text-only, without any personally identifiable information	Anonymized textual data with no personal information
UH Ethical Policies	Data anonymized under MIT license	Secondary data anonymized, collected under informed consent
Permission to Use	Used and open for research under MIT license	-
Original Data Ethics	Created By researchers from UC Berkeley, google	-

Fig. 5 Ethical check for both ETHICS and MoReBench dataset

B. Version Control

- GitHub Repository: <https://github.com/nuijnil/AI-ethics>
- Commit Frequency: Daily updates to capture progress
- Versioning System:
 - Scripts: Tracked and named using the
 - Data Storage and Versioning
 - Data predicted to be possibly larger than what standard GitHub permits.
 - Commits: Each commit with relation to stage of research (i.e. feat: preprocessing script, etc...)

C. Security & Storage

- Primary Storage: Personal Machine
- Secondary Backup: GitHub (auto-sync).
- Backup Frequency: Daily.

IV. REFERENCES

- Chiu, Y. et al., 2025. *MoReBench: Evaluating Procedural and Pluralistic Moral Reasoning in Language Models, More than Outcomes*, Online: arXiv / Cornell University Library.
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- Jiang, L. et al., 2021. Can machines learn morality? The Delphi experiment. *arXiv preprint arXiv:2110.07574*, -(-), pp. -.
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